

Threshold-Triggered Cycles as Endogenous Relaxation Experiments: Mining a Biological Methanation Rate from Quantized Pressure in a Micro-Pressurized Recirculating Reactor

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Abstract. In the final stage of anaerobic digestion, headspace pressure falls both because the biology consumes gas and because freshly injected gas dissolves; telling the two apart conventionally requires a second instrument. This paper recovers the biological rate from the pressure record alone. The reactor is a micro-pressurized recirculating vessel that refills whenever pressure falls below a controller threshold, so every refill begins an unforced descent and ordinary production supplies about two relaxation experiments a day at no cost to operation. The obstacle is that the two contributions to a single descent are nearly interchangeable, and the quantities estimated from one cycle are strongly correlated. We resolve this by calibrating the estimator against simulations that reuse each measured cycle's own fitted conditions and substitute only a known biological rate, so the distortion is measured and removed rather than assumed away. Across 239 screened cycles the calibration rules out a constant biological rate and returns a corrected consumption rate of $0.0126 \text{ kg/cm}^2/\text{hr}$, accurate to 12 percent, of which 5 percent is systematic and does not shrink with a longer record. A classifier-based two-sample test confirms that the simulator reproduces every marginal property of the measured trajectories that we tested.

Keywords: Process data mining Practical identifiability Anaerobic digestion
Quantized sensing Biomethanation

1 Introduction

The data behind this paper come from an industrial biomethanation reactor built as a cascade of pressure tiers (Fig. 1): hydrogen and carbon dioxide are premixed at high pressure, stepped down to hold the culture under a small positive pressure, and vented at low pressure to a gas analyser, while a pump continuously returns headspace gas to the bottom of the liquid as bubbles. Gas-liquid mass transfer is the limiting step of biological methanation, and holding the gases under pressure and recirculating them is how this reactor addresses it [14]. Its logger records one pressure value a minute, quantized at 0.01 kg/cm^2 . The controller that decides when to refill is a closed unit: its setpoint, its state and the timing of its decisions are never written down, nothing can be added to the instrument, and no excitation can be imposed on a vessel in production.

What the record does supply is repetition. Because refilling is threshold-triggered, every descent between two refills is a relaxation experiment the plant performs on itself, about twice a day and 280 times over the periods analysed here. Reading a

biological rate out of those descents is the measurement problem of anaerobic digestion in miniature: in a closed headspace, methanogenesis consumes five moles of gas for every one it produces and lowers pressure, while dissolution of the freshly injected CO₂ lowers it as well. The standard resolution is to add a second physical channel—manometric biomethane-potential practice, for instance, separates leakage from conversion by weighing the bottle [8]—and that option is not available here.

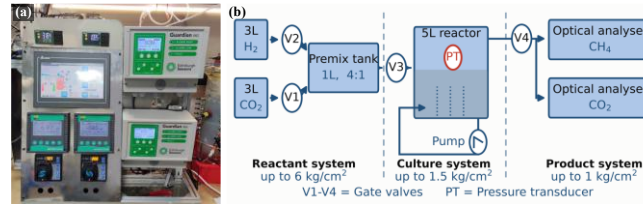


Fig. 1. (a) The instrument as built: the control panel carrying the pH and ORP meters and the two optical gas analysers, with the reactor vessel at the right. (b) The cascaded pressure-tier design of the reactor. Gas is stepped down from the reactant tier through the culture tier to the product tier, so that the culture is held under a small positive pressure while the analyser downstream stays near ambient. The recirculation pump returns headspace gas to the bottom of the vessel as bubbles, which is what keeps the reactant gases dissolving. The transducer marked PT in the headspace is the only channel this paper uses; the controller that opens valve V3 is a closed unit whose state is never logged.

The difficulty is that a single relaxation is not enough to separate the two channels. Writing the descent as

$$P(t) = P_{eq} + A e^{-kt} - r_b t \quad (1)$$

the exponential term carries the physical approach to equilibrium and the linear term the constant-rate biological removal. When kT is small over the observed window, $e^{-kt} \approx 1 - kt$ and the two terms become interchangeable. Practical identifiability of exactly this kind is a recognized open problem in the anaerobic-digestion modeling literature [4], and it persists in the mechanistic models written specifically for biomethanation: recent extensions of ADM1 to this process carry biological and mass-transfer parameters that must be identified together from the same measurements [2].

Our contributions are as follows.

- A rate where the field adds a channel instead. We treat the degeneracy head-on on a single quantized pressure channel, rather than dissolving it with a second instrument as manometric practice does [8].
- A screen that runs before the fit. Established identifiability diagnostics are computed from a fitted model; we instead reject degenerate cycles from the raw trajectory shape alone, at a threshold fixed by recovery on synthetic data at the sensor's own quantization step (Eq. (2), Algorithm 1).
- Excitation supplied by the control loop. The threshold-triggered refill makes the plant excite itself: 280 relaxation experiments with nothing imposed on production.

The statistical instruments are all established and we name their sources as we apply them; what we claim is their assembly and the regime it is made to work in.

2 Related Work

Three lines of work bear directly on this problem, and the first is the one that most constrains what we may claim.

Two rates from one gas trace. The estimation geometry we face is fifty-nine years old. The dynamic gassing-out method [1] extracts, from a single dissolved-gas trace across an interruption of aeration, both a mass-transfer coefficient and a biological uptake rate: one scalar channel, one transfer term, one biological term. We inherit that structure and claim no novelty in it. The regime differs: their protocol imposes the interruption where ours may not perturb a production vessel, and they report a point estimate where we report a degeneracy-calibrated aggregate. Dynamic pressure variants [9] likewise rely on an imposed step and treat the biological term as negligible or independently known.

Adding a channel rather than resolving the degeneracy. Where the split matters in practice, the field measures something else: manometric biomethane-potential assays separate leakage from conversion by weighing the bottle [8], a second and physically independent observable. That is why the degeneracy is rarely confronted directly. Where a quantity cannot be instrumented at all, the field increasingly infers it from the measurements already available [12]; ours is of that kind, but without labelled data to learn from, and without the option of a second channel the identifiability question is unavoidable.

Identifiability diagnostics. The standard diagnostic is the profile likelihood [11], recently reviewed together with what identifiability implies for estimation and prediction [15]; the ill-conditioning of exponential fitting was characterized by Varah [16]. Both are computed from a fit, which is the gap our screen addresses: we must decide which cycles are worth fitting before we have a fit to diagnose.

3 Reactor and Data

The Reactor

The reactor is a micro-pressurized recirculating vessel operated under a granted invention patent (TW I923176). Liquid is recirculated continuously; the headspace is charged with an $H_2:CO_2$ mixture premixed at 4:1, the stoichiometric ratio for $CO_2+4H_2\rightarrow CH_4+2H_2O$. The controller opens the inlet valve when headspace pressure falls below a threshold and closes it on reaching a target, so a refill is complete within about one minute and is treated here as instantaneous. The arrangement is that of Fig. 1; the controller reads pressure and commands the inlet valve, but its state is never logged.

The Logged Data

Of the logged channels only four are trustworthy: two pressure transducers, oxidation–reduction potential, and pH. The gas-analyzer composition channels are stale for 99.98% of samples and are not used as evidence anywhere in this paper. The present analysis uses the headspace pressure channel alone.

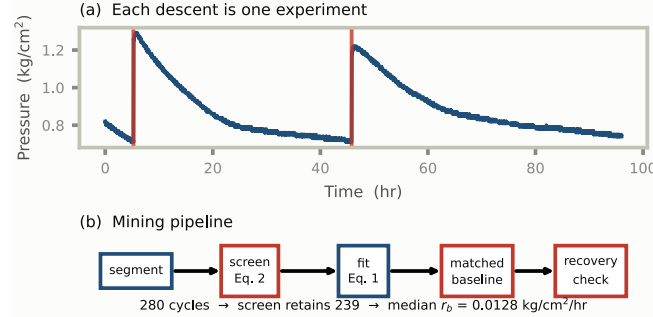


Fig. 2. (a) Four days of measured pressure; red lines mark threshold-triggered refills, about twice a day and 280 times in total. Each descending flank is one endogenous relaxation experiment. (b) The mining pipeline of Sect. 4.

Fig. 2(a) shows a representative four-day window; the sawtooth is the signature of the trigger loop. Because a refill raises pressure by far more than the sensor quantum within a single sample interval, cycle boundaries need no changepoint machinery: a threshold rule on the rise above the running minimum yields 280 cycles.

Those cycles are long and shallow. Pressure is logged once a minute, so a cycle of median duration 10.3 hr carries roughly 620 samples. The median descent amplitude is 0.31 kg/cm^2 , which is 31 quantization steps, while the median residual scale about a fitted trajectory is 0.0065 kg/cm^2 , below a single step. The biological term accounts for roughly half of the descent, so the quantity to be recovered spans on the order of fifteen steps against a per-sample noise smaller than one. That is the regime the rest of the paper works in: many samples, a coarse ladder, and an effect large against the noise but small against the physical term it must be separated from.

4 Method

Segmentation, Cleaning, and the Model Form

Fig. 3 summarises the three stages of the method: estimation, calibration, and validation.

Segmentation splits wherever pressure rises more than 0.03 kg/cm^2 above the running minimum; segments shorter than 4 hr or shallower than 0.10 kg/cm^2 are discarded, and the first 3 min of each is removed because the refill is treated as instantaneous. Cleaning removes two contaminants that the equipment operators identified: pipe-washing days, on which the vessel is cycled repeatedly to clean lines, and the refill interval itself.

The form of (1) follows from where each process acts. Methanogens consume gas that is already dissolved, so the headspace loses gas only across the interface. Just after a refill the liquid is undersaturated and transfer is fast, which gives the exponential relaxation $A e^{-kt}$; once the liquid is near saturation, transfer only replaces what the culture removes, and the headspace then drains at a rate set by the biology, which gives the linear term $-r_b t$. Treating the two as additive, rather than coupling the biological sink to the headspace, is what leaves r_b identifiable: in the coupled form $P = -k(P - P_{eq}) - r_b$ the rate enters only through the combination $P_{eq} - r_b/k$ and is absorbed

by the equilibrium pressure. Empirically, (1) wins on 78.1% of cycles by AIC against pure-linear and pure-exponential alternatives.

Its degeneracy is not a defect of our implementation but a property of exponential fitting understood since Lanczos and quantified by Varah [16]; in the language of Gutenkunst et al. [7] the model is sloppy along the direction that trades A and k against r_b . On real data two fits differing by a factor of 2.5 in r_b lie within 1% of the same residual sum of squares.

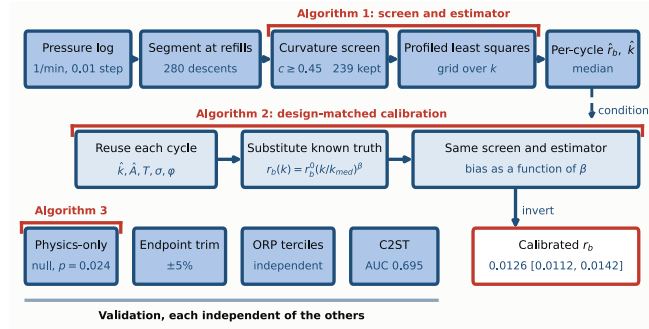


Fig. 3. The method in three stages. Top: estimation, from the raw pressure log to a per-cycle pair $(\hat{r}_b, \hat{k})$. Middle: calibration, which does not build a second pipeline but reuses each measured cycle's own fitted conditions and substitutes a known truth, so that the estimator's distortion appears on both sides of the comparison and cancels; the bias it returns is then inverted to give the calibrated rate. Bottom: four checks, each independent of the others. Brackets mark the scope of the three algorithms: Algorithm 1 is the screen together with the estimator, Algorithm 2 the design-matched calibration, and Algorithm 3 the physics-only reference.

A Pre-Fit Screen

The profile likelihood [11] would diagnose this degeneracy, but it is computed from a fit, and on this data a fit is exactly what we do not yet trust. We therefore screen on the shape of the raw trajectory before fitting anything.

The screen statistic is the normalized mid-point curvature

$$c = (y(0) - y(T/2)) / (y(0) - y(T)) \quad (2)$$

which equals exactly 0.5 for a straight descent, where the two terms of (1) are indistinguishable, and rises as the trajectory bends. A cycle is retained when $c \geq c^*$, with $c^* = 0.45$ fixed by recovery on synthetic cycles of known truth $r_b = 0.012$: below that value the recovered median is biased by more than 25%. Algorithm 1 gives the screen and the estimator; both run in NumPy alone, within the 60 MB budget of the monitoring computer.

Algorithm 1 Extracting a Trustworthy Rate from One Cycle

Input: a single relaxation cycle (t, y) of duration T

Parameter: curvature threshold $c^* = 0.45$, grid $K = \{0.01, 0.03, \dots, 3.00\}$

Output: the estimates r_b and k for this cycle

- 1: $c \leftarrow (y(0) - y(T/2)) / (y(0) - y(T))$ // shape only; no fit needed
- 2: **if** $c < c^*$ **then reject the cycle and stop**
- 3: **for each** $k \in K$ **do**

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4:   D(k) ← [exp(-k t), t, 1]      // design matrix
5:   b(k) ← argminv ||y - D(k) v||2 // 3 × 3 normal equations
6:   s(k) ← ||y - D(k) b(k)||2    // residual sum of squares
7:   end for
8:   kfit ← argmink s(k)
9:   return rb = -b2(kfit) and kfit

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[Cycles too straight to constrain the relaxation constant are rejected before any fit is attempted. The profiled least-squares estimator for (1). Solving all 150 grid values of k as one stacked 3×3 system keeps a full pass over 280 cycles under one second]

Calibrating the Degeneracy Away

Across the retained cycles the fitted $\hat{r}_b$ and $\hat{k}$ correlate at $\rho=0.814$, which read naively looks like proof that the estimate is an artifact. It is not readable in isolation: $\hat{r}_b$ and $\hat{k}$ jointly absorb the same descent, so they correlate even when the truth is fixed. The question is how far the correlation exceeds what a fixed truth already produces.

We therefore apply empirical calibration in the sense of Schuemie et al. [13], replacing the theoretical null $\rho=0$ with one estimated from cases where the truth is known. The null is generated by design-matched simulation (Algorithm 2): each real cycle contributes its own fitted $\hat{k}, \hat{A}, \hat{P}_{eq}$, duration, residual scale and AR(1) coefficient, only the true r_b is substituted, and the result passes through the identical pipeline, so the estimator's distortion appears on both sides and cancels—the logic of indirect inference [6]. Our adaptation is the source of the negative controls: ours are simulations reusing the real designs, which fixes the truth exactly instead of assuming it.

Algorithm 2 Measuring and Removing the Degeneracy Bias

Input: per-cycle fitted designs $k_i, A_i, P_{eq,i}, t_i, \sigma_{\epsilon,i}, \phi_i$
Parameter: candidate truth $r_b(\cdot)$, repetitions $R = 25$, quantum $q = 0.01$
Output: baseline distributions of ρ and m

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1:   for r ← 1 to R do
2:     for i ← 1 to 239 do
3:       e ← AR(1) noise, coefficient 1.3  $\phi_i$ , scale  $\sigma_{\epsilon,i}$ 
4:       z ←  $P_{eq,i} + A_i \exp(-k_i t_i) - r_b(k_i) t_i + e$ 
5:        $y_i \leftarrow q \text{ round}(z / q)$  // identical quantization
6:        $(r_{b,i}, k_{k,i}) \leftarrow \text{Algorithm 1}(t_i, y_i)$ 
7:     end for
8:      $\rho(r) \leftarrow \text{Spearman}(r_b, k_k)$  // over the 239 cycles
9:      $m(r) \leftarrow \text{median}(r_b)$ 
10:  end for
11:  return the distributions of  $\rho$  and  $m$ 

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[Sweeping β with this routine gives the curve of Fig. 4; a constant truth ($\beta=0$) returns $\rho=0.304 \pm 0.056$, and a truth independent of k returns $\rho=0.143 \pm 0.048$, against an observed 0.814]

The simulator itself must be shown adequate, or the baseline means nothing. We use a classifier two-sample test, introduced for this purpose by Friedman [5] and revisited by Lopez-Paz and Oquab [10], and applied to simulator and emulator validation by Dalmasso et al. [3]: a classifier is trained to tell real trajectories from simulated ones, and an AUC near 0.5 indicates the two are indistinguishable. The noise model is calibrated on one half of the cycles and the AUC computed on the held-out half.

5 Results

The Rate

The screen retains 239 of the 280 cycles. Over these the median r_b is $0.0128 \text{ kg/cm}^2/\text{hr}$ (the calibrated rate). Against a misspecification null in which the truth is a pure exponential, that is, $r_b = 0$ and all pressure loss physical, the observed median is reached with permutation $p=0.024$. The pressure descent therefore contains a component that a physics-only model cannot produce. We stress the direction of this inference: it excludes a purely physical explanation; it does not by itself establish that the residual component is biological.

What the Correlation Means

Rather than assume a form for the k dependence, we treat its strength as the quantity to be inverted. Writing the truth as $r_b(k) = r_b^0 (k/k_{med})^\beta$ —a power law, so that r_b cannot turn negative and the median truth is r_b^0 for every β —we sweep β and run the full matched pipeline at each value, obtaining the calibration curve of Fig. 4(a).

Table 1. The recovered rate and the calibration checks behind it. The baseline rows are design-matched simulations in which the true rate is known, so the estimator's own distortion can be read off directly.

Quantity	Value	Note
Cycles segmented	280	threshold rule
Cycles retained	239	$c \geq 0.45$
Calibrated r_b	$0.0126 \text{ kg/cm}^2/\text{hr}$	95% [0.0112, 0.0142]
sampling component	$\pm 5.5\%$	bootstrap over 239 cycles
systematic component	$\pm 5.1\%$	over the compatible set
Physics-only reference	$p=0.024$	pure exponential truth
Observed $\rho(\hat{r}_b, \hat{k})$	0.814	
Calibration, $\beta=0$ (constant)	0.304 ± 0.056	artefact level
Baseline, r_b independent of k	0.143 ± 0.048	
Compatible set	$0.20 \leq \beta \leq 0.80$	within calibration scatter
Raw profiled median	$0.0128 \text{ kg/cm}^2/\text{hr}$	before correction
Calibrated correction	-3.7%	estimator distortion only

A constant truth is excluded: $\beta=0$ produces only $\rho=0.304 \pm 0.056$ against the observed 0.814, a gap of roughly nine standard deviations, and making r_b merely variable without tying it to k lowers the correlation further, to 0.143. Truths with β between 0.20 and 0.80 reproduce the observed correlation to within the calibration scatter, and it is that compatible set we carry forward.

The same calibration then supplies the correction (Fig. 4(b)), and it separates two effects that must not be conflated. The screen admits cycles by curvature, so the truths it keeps are not those of the whole population; that displacement grows with β and reaches 37% at the top of the compatible set. It is not an error but a definition: we report the rate on the cycles where it is identifiable, and the screen is what makes them so. What must be corrected is the estimator's own distortion, measured against the truths actually kept, and that stays between -4.1% and +5.8% across the whole compatible set. Inverting it gives the calibrated rate

$$r_b = 0.0126 [0.0112, 0.0142] \text{ kg/cm}^2/\text{hr} \quad (3)$$

a correction of -3.7% on the raw profiled median.

The interval combines two sources that behave differently. Resampling the 239 cycles gives $\pm 5.5\%$ on the median, which shrinks as logging continues; the spread over the compatible truths is $\pm 5.1\%$ and does not, being set by how sharply the calibration separates members of the family rather than by how many cycles are seen. Since the correction enters as a quotient we draw from both by Monte Carlo, giving the $\pm 11.9\%$ of (3)—the accuracy with which a single quantized pressure channel fixes the biological rate, and the main quantitative result of this paper.

How the Reference Distribution Is Built

The validity of the test behind $p=0.024$ (Algorithm 3) rests on what the null is allowed to contain. For each retained cycle we fit the two-parameter form $P(t)=P_{eq}+A e^{-kt}$, that is (1) with r_b forced to zero, so the fitted null reproduces the cycle's own equilibrium, amplitude and relaxation constant. We regenerate the cycle from that fit, add AR(1) noise at its own residual scale, quantize to 0.01 kg/cm^2 , and pass it through the identical screen and estimator. Over all 239 cycles this yields one draw of the aggregate median under a physics-only truth; 60 draws give its null distribution.

Algorithm 3 Ruling Out a Purely Physical Explanation

Input: the retained cycles (t_i, y_i) , $i = 1, \dots, 239$
Parameter: draws $D = 60$, grid K , sensor quantum $q = 0.01$
Output: the reference distribution of the aggregate median

```

1:   for  $i \leftarrow 1$  to 239 do
2:     fit  $P_{eq,i} + A_i \exp(-k_i t)$  to  $(t_i, y_i)$  // rate forced to zero
3:      $\text{sigma}_i \leftarrow$  residual scale of that fit
4:   end for
5:   for  $d \leftarrow 1$  to  $D$  do
6:     for  $i \leftarrow 1$  to 239 do
7:        $e \leftarrow$  AR(1) noise at scale  $\text{sigma}_i$ 
8:        $z \leftarrow P_{eq,i} + A_i \exp(-k_i t_i) + e$ 
9:        $w_i \leftarrow q \text{ round}(z / q)$  // identical quantization
10:       $rb_i \leftarrow$  Algorithm 1  $(t_i, w_i)$  // identical estimator
11:    end for
12:     $m(d) \leftarrow \text{median}(rb)$ 
13:  end for
14:  return the distribution of  $m$  over the  $D$  draws

```

[The reference is fitted per cycle, so it inherits every degree of freedom the alternative has except the biological rate itself, and it passes through the same quantization, screen and estimator]

The null is fitted per cycle rather than pooled, so it is granted every degree of freedom the alternative has except r_b itself. Under it the aggregate median is $+0.00008 \pm 0.00010 \text{ kg/cm}^2/\text{hr}$, two orders of magnitude below the observed 0.0128 .

Robustness to the Trigger-Selected Endpoint

Segmentation ends each cycle at the sample immediately before a refill, and that sample is selected because pressure fell far enough to trigger, so it sits below the underlying trend: stacked residuals give a terminal mean of -0.068 of the cycle

amplitude against ± 0.006 elsewhere. Since a linear slope is most sensitive at its endpoints, this is a plausible route to an inflated rate.

Trimming settles it. Removing the last 5 to 90 minutes of every cycle moves the median r_b only within $0.0124\text{--}0.0133\text{ kg/cm}^2/\text{hr}$, a band of $\pm 5\%$, and a design-matched simulation with no trigger selection drifts comparably: one anomalous sample among roughly 620 has too little leverage to bend the slope.

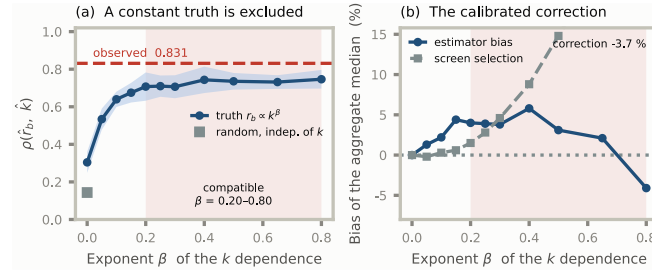


Fig. 4. (a) Calibration of the correlation against the strength β of the k dependence, each point a full matched pipeline; the band is ± 1 standard deviation over repetitions and the shaded region is the compatible set. A constant truth ($\beta=0$) falls nine standard deviations below the observation. (b) The bias read from the same calibration: the estimator's own distortion, which is what we correct, and the screen's selection of truths, which is definitional; inverting it yields the -3.7% correction applied to the rate.

An Independent Biological Check

Everything above is derived from pressure, and surviving a physics-only reference says what the residual component cannot be, not what it is. The reactor also logs oxidation–reduction potential. It enters no part of the estimate and is not a rate: it responds to the reducing power of the dissolved hydrogen, indexing how active the culture is rather than how fast gas disappears, which is what makes it useful as a check.

Sorting the cycles by the redox excursion each one shows and comparing the aggregate rate across terciles gives 0.0099 , 0.0134 and $0.0152\text{ kg/cm}^2/\text{hr}$ from least to most active, a 54% span whose bootstrap interval, $[0.0004, 0.0137]$, excludes zero. A quantity measured on a different physical channel, in the liquid rather than the gas, therefore orders the same rates that the pressure trajectories give.

The ordering is aggregate rather than per cycle, and what it establishes is that the variation the calibration detected in r_b tracks an independently measured signature of biological activity, which pressure alone could not have shown.

Simulator Adequacy

The classifier two-sample test is reported next. A white-Gaussian noise model is separated at $AUC=0.968$; the residuals of real cycles in fact carry a lag-1 autocorrelation with median 0.266 . Modeling the noise as AR(1) reduces this to 0.832 , and inflating $\hat{\phi}$ by the factor 1.3 calibrated on the held-out split reaches 0.695 . All ten marginal trajectory features then agree to within 0.29 pooled standard deviations, the largest gap being the monotone fraction (0.808 against 0.786), with

curvature, residual scale, lag-1 autocorrelation, amplitude and duration all within 0.15; we return to the residual mismatch in Sect. 6.

6 Limitations

Three limits bound what the preceding numbers support. First, the aggregate median is recoverable but the per-cycle estimates are not: individual values scatter widely and are coupled to $\hat{k}$, so nothing here licenses reading a single cycle as a rate measurement. The same applies to the redox check: it also responds to pH and to dissolved carbon dioxide, so it is not a hydrogen probe, and without a composition anchor it cannot be turned into a rate to set against (3). Of the $\pm 11.9\%$ on the rate, the $\pm 5.1\%$ systematic part is fixed by the calibration rather than by the record length; narrowing it needs the form of $r_b(k)$ itself, which a second chemical channel would supply.

Second, the simulator remains distinguishable from reality at $AUC=0.695$ even after all ten marginal features are matched, so the mismatch lies in joint structure we have not identified. The matched baselines are consequently approximate, and we do not attempt simulation-based inference on this simulator: a posterior learned from a misspecified simulator would be narrow and wrong rather than merely uncertain.

Third, the deployment target is a monitoring computer with 4 GB of RAM and roughly 60 MB available, where the binding constraint is library footprint rather than algorithmic cost—NumPy alone occupies 27 MB against 129 MB for the full scientific stack—so the pipeline is written against NumPy only. That is also why we did not pursue a learned model: the difficulty is identifiability rather than function approximation, and extra parameters would absorb the same degeneracy with less visibility.

7 Conclusion

A threshold-triggered micro-pressurized reactor performs relaxation experiments on itself, and those experiments carry a recoverable biological rate even when the only observable is a single pressure channel quantized at 0.01 kg/cm^2 . Reading that channel correctly requires calibrating the estimator against a design-matched null rather than a theoretical one, because the raw correlation between the two fitted terms is uninterpretable on its own. Adding a second channel—a composition anchor or an improved redox signal—is the direct route to a per-cycle result.

Acknowledgements. The reactor is operated under invention patent TW I923176 held by the Metal Industries Research & Development Centre.

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